

YANG XINYANG

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EDUCATION

National University of Singapore (NUS)

2024 – Present

Ph.D. in Computer Science, Research Scholarship

School of Computing

Shanghai Jiao Tong University (SJTU)

2020 – 2024

Bachelor of Engineering in Computer Science

Member of ACM Class, a selective CS honors program for the top 5% of students

RESEARCH EXPERIENCE

School of Computing, National University of Singapore (NUS)

Supervised by *Prof. Jiaheng Zhang*

Aug 2024 – Present

- Conduct research on efficient zero-knowledge proof systems, with a focus on succinct range proofs and verifiable approximate nearest-neighbor retrieval.

School of Computing, National University of Singapore (NUS)

Supervised by *Prof. Yu Chen*

Dec 2024 – Present

- Study the multicommodity flow-cut gap in directed graphs, with a particular focus on directed Okamura–Seymour graphs.

School of Computer Science, Carnegie Mellon University (CMU)

Supervised by *Prof. Richard Peng*

Sep 2023 – Feb 2024

- Studied applications of analytic methods to graph data structures and algorithms, focusing on variants of multiplicative-weights update methods for approximately solving undirected multicommodity flow problems.

Basic Studies in Computing Science (BASICS Lab), SJTU

Supervised by *Prof. Yijia Chen*

Jul 2022 – Jul 2023

- Conducted research in computational complexity and algorithmic graph theory, focusing on forbidden-subgraph theorems and DPLL-based algorithms.

PUBLICATIONS AND MANUSCRIPTS

- **Rarus: A Succinct and Efficient Range Proof for Polynomial-Based Vector Commitments.** First author. Accepted to the 35th USENIX Security Symposium (USENIX Security '26).
- **zkRAG: Efficiently Proving RAG Retrieval in Zero Knowledge.** Co-first author. Manuscript submitted to USENIX Security 2027 Cycle 1.

SELECTED PROJECTS

BFBT: Binance Futures Backtesting Framework

Independent Open-Source Project — *Interactive Reports*

Aug 2026 – Present

- Built a local-first research system for cross-sectional factors on Binance USD-M perpetual futures, with a three-stage pipeline spanning Quick Research diagnostics, Fast Matrix portfolio evaluation, and formal event-driven backtesting.
- Engineered point-in-time data preparation, explicit fees, slippage and funding, path-dependent risk controls, bounded-memory checkpoint recovery, and immutable data, configuration, code and run identities for reproducible experiments.
- Added a model-agnostic, supervised AI Agent workflow that translates natural-language research requests into validated, authorization-gated stages and evidence-cited results.

PromptPerp: AI-Native Binance Futures Trading Infrastructure

Independent Open-Source Project

Oct 2026 – Present

- Built a safety-gated framework that translates natural-language strategy ideas into validated, approval-bound candidates for Binance USD-M futures.
- Implemented recoverable order execution, risk controls, account reconciliation, and investor accounting.

HONORS AND AWARDS

Excellence Scholarship for Undergraduates

2021, 2022, 2023

Zhiyuan Honorary Scholarship, awarded to the top 5% of students

2020, 2021, 2022, 2023

SKILLS

Cryptography: Applied Cryptography, Zero-Knowledge Proofs, Polynomial Commitments

Programming: C/C++, Python, Java, RISC-V Assembly

Quantitative Research: Cross-Sectional Factors, Portfolio Research, Event-Driven Backtesting, Reproducible Experimentation

Hardware and Systems: Verilog

Mathematics: Linear Algebra, Real Analysis, Graph Theory, Combinatorics, Complexity Theory